HAMR 3.0: Off-Road Holonomy, Traversability, and Control

Fall 2025 Independent Study Progress Report

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I. Introduction

This report summarizes the work completed during the Fall 2025 semester as part of my independent study on HAMR 3.0, an off-road evolution of the holonomic differential-drive platform originally introduced in HAMR. The project objective was to extend holonomic capability to rugged terrain through realistic simulation, traversability reasoning, advanced control design and new hardware.

The work was conducted in collaboration with George Mason University (GMU), whose COMPA platform provided a strong off-road mechanical base. Our contributions focused on simulation, controls, traversability analysis, software architecture, and experimental validation.

This report documents:

- Simulation and terrain environment development
- Hardware evolution and team execution
- Vicon-based validation and experimentation
- 2.5D traversability analysis and planning
- Transition from PID controllers to MPC-based control
- Documentation and ROS 2 system improvements
- Current status and forward roadmap

II. Background and System Context

HAMR 3.0 builds upon the original HAMR concept [1] of a differential drive platform with an offset turret capable of achieving holonomic motion at the payload level, but it extends this core idea significantly by aiming to operate in meaningful off-road environments. The foundation of this semester's work was heavily influenced by the collaboration with GMU, whose COMPA platform demonstrated that rocker suspension, turret stabilization, and off-road traction could coexist within the DDROT architecture. This gave us a robust grounding in real-world constraints and reinforced that the system objective is not only to be holonomic, but to do so while preserving turret stability, payload usability, and terrain adaptability in the presence of uneven slopes and roughness variations. The evolution of HAMR 2.0 from this summer to COMPA to HAMR 3.0 can be seen in Figure 1. HAMR 3.0 attempts to connect theoretical holonomicity tools – like the mobility ellipsoid of COMPA (Fig. 2) we re-explored in our paper submitted to ICRA 2026 – with 2.5D traversability planning approaches that reason about terrain geometry and feed given constraints into the control stack.

Fig. 1: Hardware evolution – Penn HAMR 2.0, GMU COMPA, and Penn HAMR 3.0

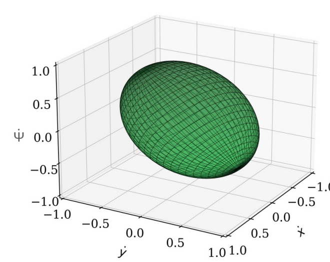

Fig. 2: Projected ME: $\dot{x}$, $\dot{y}$, and $\dot{\psi}$ of COMPA.

From the perspective of the original grading rubric, the "theoretical analysis of HAMR" was initially envisioned as a mathematical deep dive into the holonomic properties of the DDROT structure and its Jacobian formulation across different terrain conditions. Over the semester, however, it became clear that a much more impactful and practically valuable direction was to redirect this theoretical focus toward developing a full **traversability reasoning pipeline**. Although this deviates from the original phrasing of the rubric, it does not deviate in spirit; instead, it strengthens it. Rather than performing theoretical analysis in isolation, we pursued theoretical work that directly informs navigation, autonomy, and control. This shift is academically justified, provides direct impact on system capability, and prepares HAMR 3.0 as a genuinely useful off-road platform rather than a lab prototype.

III. Experimental Validation: Vicon Testing

A significant portion of this semester involved empirical testing of the existing HAMR-based system using the Vicon motion capture system at Pennovation Perch (Fig. 3). This allowed us to rigorously evaluate whether controllers behaved in the real world as expected from simulation and to identify structural and mechanical limitations inherent to the previous generation platform. The motion capture experiments provided highly accurate ground truth pose tracking, enabling comparison with onboard odometry Fig. 4). This confirmed that our estimation pipeline, based on motor encoders and IMU readings, was functioning correctly and provided confidence that future autonomy stacks would have a credible state representation to rely upon.

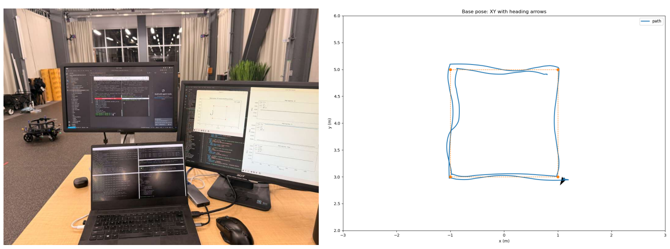

Fig. 3: Vicon experimental setup and robot testing.

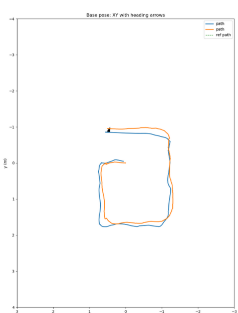

Fig. 4: Odometry validation: agreement between estimated and ground truth trajectories.

At the same time, these experiments also exposed one of the most important physical insights of the semester: the swivel caster architecture introduced oscillations, drift, and undesirable dynamic coupling, particularly when commanding nominally straight trajectories. Several experiments where straight-line motion was commanded demonstrated lateral oscillations and steering drift that were not controller artifacts, but mechanical effects. This discovery strongly motivated the full mechanical redesign and justified the allocation of team resources toward building a new platform, specifically focusing on designing new off-road adapted casters. Importantly for the grading rubric, these experiments fulfilled and exceeded the expectations of "physical testing and validation." Even though full outdoor field deployment was not yet reached due to hardware redesign timelines, the Vicon work provided essential data-driven justification for redesign and contributed directly to the trajectory of the project.

Additionally, COMPA's performance reported in our submitted paper (Table I) demonstrates strong path following consistency on canonical paths such as squares, triangles, and circles. The RMSE reported validates that, even with existing hardware imperfections, the holonomic framework performs competitively and predictably.

TABLE I: Physical Experiments Results - COMPA.

Path	**Path RMSE** (m)	**Roll Std.** (rad)	**Pitch Std.** (rad)	**Yaw Std.** (rad)
Square	0.1776	**0.0439**	**0.0404**	0.1067
Triangle	0.1973	0.0447	0.0441	0.1597
Circle	**0.1005**	0.0457	0.0438	**0.0501**

IV. Literature Review on Traversability and Planning

A substantial portion of this independent study was dedicated to understanding how existing work treats traversability, elevation mapping, and planning on rough terrain. Over the semester I went through more than twenty papers, organized in a shared reading log and weekly slide decks, and tried to extract what would be directly useful for HAMR/COMPA versus what should remain as future inspiration.

At a high level, the survey by Sevastopoulos and Konstantopoulos [2] gave me a taxonomy of traversability estimation approaches (geometric, physics-based, learning-based) and clarified where my work fits: a primarily geometric/physics-inspired 2.5D pipeline with the possibility of adding learned components later. Fankhauser and Hutter's grid map chapter [3] complemented that by providing the concrete data structures and ROS tooling that I actually used (layers for elevation, slope, roughness, etc.), which directly motivated my decision to implement traversability as a new `grid_map` filter layer rather than invent a custom format. I also iterated some of their files for our own use case.

Several works focused specifically on 2.5D terrain representations and classical traversability indices. Gu's rapid 2.5D assessment [4] and 2.5D path planning paper [5] showed how simple geometric features such as local slope and step height can be encoded per-cell and then fed to a grid-based planner. Wang et al. [6] extended this idea on rigid terrain with a more articulated cost pipeline and explicit vehicle constraints, which was conceptually close to what I wanted for our robot. In parallel, Fankhauser et al. [7] introduced probabilistic `elevation_mapping` with uncertain localization, which I did not implement this semester but which clarified how elevation uncertainty could later be folded into the cost field.

We will most probably use the modernized ROS 2 version called `[elevation_mapping_cupy]`. Waibel et al. [8] further influenced my thinking by treating "roughness" as an explicit signal for both local and global planning, reinforcing my decision to keep a secondary roughness term in the traversability cost even though slope is dominant for COMPA.

Other papers explored richer, more physically grounded notions of difficulty. Carvalho et al. [9] defined traversability in terms of mechanical effort in forest environments, while Eder et al. [10] combined locomotion experiments with earth-observation data to learn cost fields at scale. Goulet and Ayalew [11] optimized trajectories on deformable terrain with an explicit energy cost-to-go coupled to MPC. I did not reproduce those models, but they pushed me to think about COMPA's cost function as something that should encode actual mobility limits (energy, slip risk, rollover risk) instead of arbitrary numbers. This perspective is reflected in the way slope limits and rocker performance are embedded in my exponential traversability cost.

On the mapping and learning side, Prinz et al. [12] built off-road navigation maps from aerial imagery using convolutional networks, while Pan et al. GMU's TNT estimator [13] framed traversability for vertically challenging terrain as a data-driven function of past kinodynamic interactions. PUTN [14] combined plane-fitting, Gaussian process regression, and NMPC to navigate highly uneven terrain. I treated these works mainly as forward-looking references: they showed how my current 2.5D geometric approach could eventually be extended with learned priors (from satellite or onboard imagery) or more sophisticated local dynamics models, but they were beyond the scope of this semester's implementation.

The planning literature I read informed both the global ARA*/A* design and the MPC roadmap. Dolgov et al. [15] and Chi et al. [16] describe practical search and Hybrid A* variants for car-like vehicles, emphasising heuristics, cost inflation and repair that inspired my choice of ARA* and the notion of trading optimality for speed. Xu et al. [17] and Schichler et al. [18] showed how motion primitives, curvature constraints and smoothing can be layered on top of grid search, which is conceptually similar to how my local MPC refines the global ARA* path rather than following it blindly. Osmankovic et al. [19] provided a concrete example of coupling D* Lite with MPC for off-road vehicles; even though COMPA's MPC is not fully deployed yet, their architecture helped justify my decision to move from pure PID control toward a predictive scheme that can encode terrain costs and holonomic constraints directly.

Finally, several works explored alternative global planning back-ends or specialized traversability metrics. Garrido et al. [20] applied anisotropic fast marching for Mars rover path planning, which I considered as an alternative to ARA* on continuous cost fields. Karakaya [21] proposed slope-aware motion planning, explicitly penalizing steep climbs, which aligned well with my decision to make slope the primary term in COMPA's traversability score. The survey and off-road studies I read also referenced many follow-on ideas I noted but did not implement, such as roughness-based comfort metrics and hierarchical route guidance for long-range missions.

Overall, this reading block gave me two concrete outcomes. First, it justified the specific structure of my traversability cost: a physically motivated, slope-dominant term with secondary roughness penalties, implemented on top of `grid_map` and consumable by ARA* and MPC. Second, it clarified which directions are most promising for future extensions of HAMR/COMPA: probabilistic and learned terrain mapping, energy- or mechanical-effort-aware costs, and tighter couplings between global search and predictive local control.

V. Computer Vision and Terrain Reconstruction

One of the clearest divergences from the initial rubric, and one important achievement this semester, was the beginning of a perception-driven traversability framework. Instead of performing only a "theoretical HAMR platform analysis," I redirected some effort toward developing the foundations of a computer vision and terrain reconstruction pipeline that will later enable real-time, perception-informed navigation. Using live data from an Intel RealSense camera, we demonstrated that we can generate meaningful elevation representations of the environment and convert them into ROS 2 grid map layers. Although this work was not as extensive as other components of the project, the results serve as a strong proof of feasibility and clearly establish a path toward integrating perception, traversability reasoning, and control.

The results achieved by Kartik, shown in Fig. 5, demonstrate live Structure from Motion reconstruction from the RealSense camera, producing dense environmental geometry. Meanwhile, Fig. 6 shows how we processed accumulated PCL point clouds into consistent elevation and traversability grid maps in RViz, both in controlled environments and using recorded datasets. These results confirm that the platform can move beyond predefined terrain assumptions and begin grounding autonomy in sensed world structure. Conceptually, this direction still fits the intent of the rubric, because rather than treating elevation as a theoretical construct, it becomes an operational constraint and an input into the optimization and control framework discussed later.

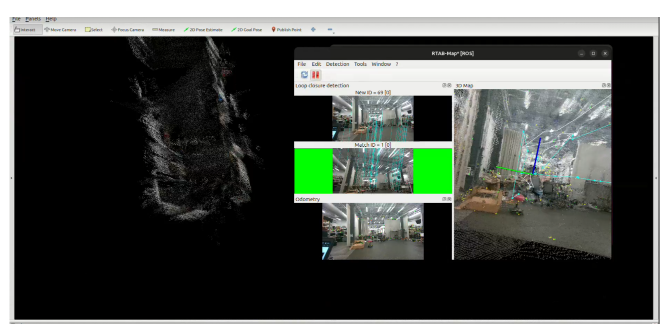

Fig. 5: Live Structure-from-Motion (SfM) reconstruction from Intel RealSense camera.

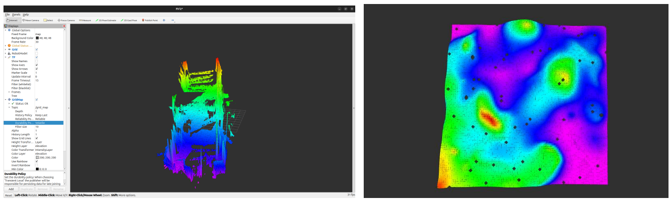

Fig. 6: Elevation and traversability grid maps generated from PCL reconstruction in ROS 2.

VI. Simulation and Terrain Development

A major outcome of this semester was the development of a high-fidelity Gazebo simulation environment (Fig. 7) explicitly designed to capture the physics, geometry, and uncertainty of off-road conditions. Very early in the semester, before new hardware progress began, a full simulation of the original HAMR and COMPA platforms were constructed to support algorithm prototyping. This environment gradually evolved to incorporate realistic 2.5D digital elevation maps, representing terrains comparable to rocky hillside environments. These terrains were fed into RViz via the `grid_map` library (Fig. 8), allowing height data to be filtered into meaningful terrain metrics such as slope and roughness, which later became fundamental building blocks of the traversability framework. We then evaluated the holonomic behavior of canonical trajectories in the simulation (Fig. 9), and compared them to Table I's real-world results with GMU's COMPA. The progression of simulation capabilities followed a meaningful arc: first, building a digital twin capable of baseline mobility; second, adding terrain fidelity; and third, moving toward autonomy through integration with ARA* planning and optimization control.

Relative to the original independent study rubric, this portion corresponds very closely to the "simulation environment development" and partially to the "rigorous analysis" expectation. These choices ensured that the work remained research-facing while still being structured and results-driven. Overall, this semester produced not only a functioning simulation stack, but one deeply aligned with the long-term off-road ambitions of HAMR.

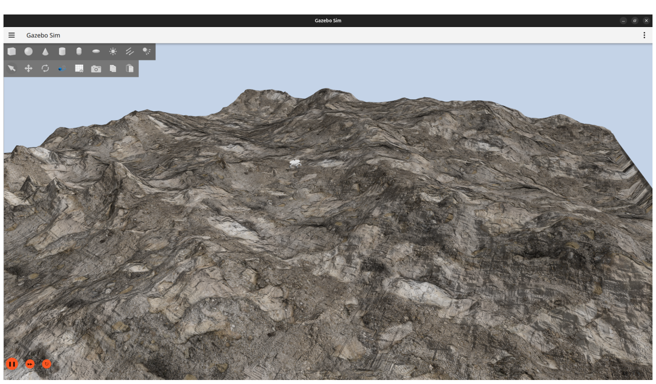

Fig. 7: Off-Road map in Gazebo.

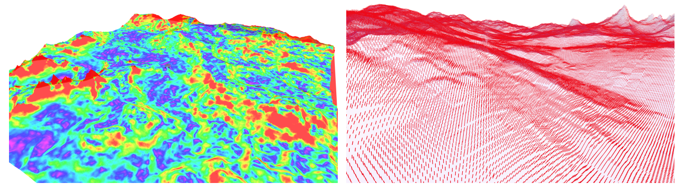

Fig. 8: Off-Road traversability analysis in RViz and surface normals using grid_map.

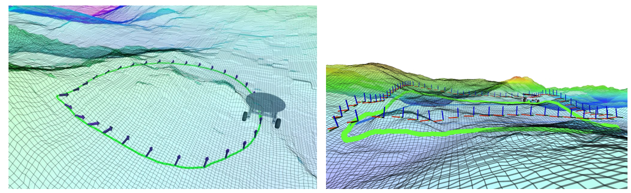

Fig. 9: Simulated holonomic canonical trajectories executed on rough terrain.

VII. Planning and Control

The planning and control development this semester formed the intellectual bridge between perception-driven traversability work and the platform's holonomic motion capabilities. Globally, I implemented ARA* [22] in C++ as the global planning framework, chosen because of its ability to trade optimality for computation speed and gradually refine solutions as we explore more cells or slightly distance ourselves from the reference – which is very likely when using a smarter local planner. Figure 10 highlights ARA* in action, where the green area represents all the explored cells. We can see in the left figure near the goal how ARA* recomputes several paths until it finds the most optimal one within the max number of time steps. It is also able to repair paths when the environment changes (helpful once we switch to elevation maps gathered live from PCL) or if the robot's trajectory differs significantly from the provided reference. The global traversability map, derived from the slope and roughness layers, served as the basis for the cost structure. This provided a mathematically meaningful global policy: lower slope regions are rewarded, and roughness introduces additional penalties, making the planner aggressively prefer safer terrain while still allowing feasible paths when the environment is constrained. This behavior is formally captured by the cost function in (1), which is implemented as a dedicated layer within a filter chain and evaluated through the `grid_map` framework:

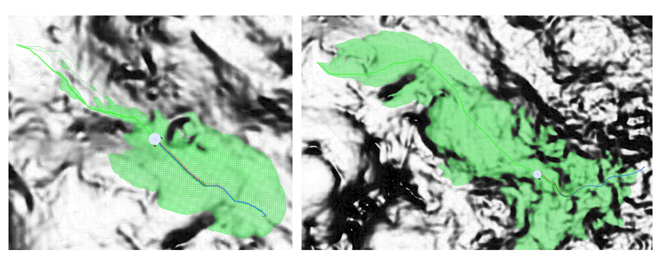

Fig. 10: ARA* global traversability planning.

$$T = w_{s,1} \exp\left(-\left(\frac{\text{slope}}{w_{s,2}}\right)^4\right) + w_{r,1} \exp\left(-\left(\frac{\text{roughness}}{w_{r,2}}\right)^4\right) \tag{1}$$

To ground this cost formulation in physically meaningful constraints, we first analyzed COMPA's slope capability in simulation and using insights from related work. As illustrated in Fig. 11, we empirically observed that COMPA can reliably operate up to approximately 0.5 rad ($\approx 30°$), but performance degrades sharply beyond this point. This value was therefore selected as the reference "maximum safe slope" threshold for the platform. Importantly, we also found that COMPA's rocker suspension allows it to handle moderate roughness surprisingly well, whereas slope induces a much clearer, dominant limitation on stability and mobility. This justified weighting slope more heavily than roughness in the traversability formulation, leading to our chosen values of $w_{s,1} = 0.9$ and $w_{r,1} = 0.1$.

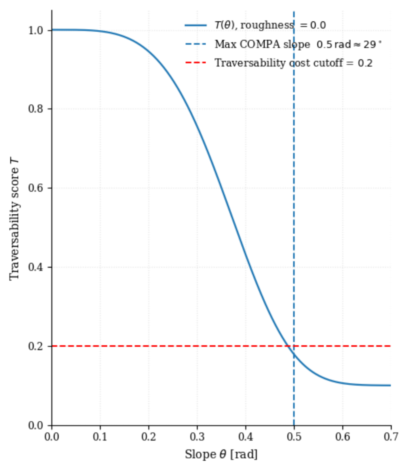

Fig. 11: Traversability w.r.t slope when roughness is set to 0.

The exponential structure of the traversability function was tuned accordingly to reflect this observed behavior. In particular, a fourth-power decay term was selected so that the traversability score drops sharply as the commanded path approaches the platform's maximum traversable slope. The parameter $w_{s,2} = 0.4$ was then chosen to shape this transition such that the traversability value approaches zero near the 0.5 rad limit with a roughness of 0.0, effectively encoding COMPA's empirically observed capability into the cost field. This tuning aligns both with our simulation results and with reported observations in related literature [Paper Ref].

Similarly, the roughness scaling parameter was tuned to reflect the fact that COMPA's rocker suspension tolerates moderate terrain irregularities while still being penalized on highly discontinuous ground. The value $w_{r,2} = 0.03$ was therefore selected so that roughness penalties remain relatively mild for small variations but begin to increase exponentially once surface disturbances exceed approximately $3\,\text{cm}^2$ per grid cell. As shown in Fig. 11, this formulation produces a steep nonlinear drop in traversability near the maximum operational slope, capturing the intuition that terrain rapidly transitions from "manageable" to "dangerous." Lower-order polynomial exponents such as 2 or 3 were evaluated but did not sufficiently emphasize this sharp transition, which ultimately motivated the adoption of the fourth-power exponential form.

Locally, the controller roadmap evolved significantly over the semester. The rubric originally called for "rigorous holonomic control and gimbal stabilization," and in practice this manifested as a staged evolution. Initially, we implemented a PID-based Cartesian control framework, which worked well and validated basic holonomic motion both in simulation and hardware experiments. However, as the ambition shifted toward off-road reliability, this transitioned toward a Model Predictive Control (MPC) formulation. The MPC objective is able to jointly reason over path-tracking accuracy, control effort, and terrain-dependent stability costs:

$$J = \sum_{k=0}^{N-1} \left(\|x_k - x_k^{\text{ref}}\|_Q + \|u_k\|_R + c_{\text{terrain}}(x_k) \right) + \|x_N - x_N^{\text{ref}}\|_P \tag{2}$$

where x_k and u_k denote the system state and control inputs at step k, and $c_{\text{terrain}}(x_k)$ encodes terrain-dependent stability penalties along the MPC horizon. To make this term physically meaningful, we approximate the base roll ϕ_k and pitch θ_k at each candidate state x_k by querying the local elevation map at the projected wheel contact points [13].

We define the wheel contact locations in the body frame as

$$\mathbf{p}_{FL} = \begin{bmatrix} +L/2 \\ +W_f/2 \end{bmatrix}, \quad \mathbf{p}_{FR} = \begin{bmatrix} +L/2 \\ -W_f/2 \end{bmatrix},$$
$$\mathbf{p}_{RL} = \begin{bmatrix} -L/2 \\ +W_r/2 \end{bmatrix}, \quad \mathbf{p}_{RR} = \begin{bmatrix} -L/2 \\ -W_r/2 \end{bmatrix},$$

where L is the wheelbase, and W_f, W_r are the front and rear tracks, respectively. For a base pose (x, y, ψ) extracted from x_k, the corresponding wheel positions in the world frame are

$$\mathbf{p}_i^{\text{world}} = \begin{bmatrix} x \\ y \end{bmatrix} + R(\psi)\,\mathbf{p}_i, \quad R(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix},$$

and the local wheel heights are obtained by bilinear interpolation on the `elevation_local` grid:

$$h_i = h\big(\mathbf{p}_i^{\text{world}}\big).$$

Using these heights, we approximate the base roll and pitch as

$$\phi_k \approx \frac{1}{2}\left(\arctan\frac{h_{FL} - h_{FR}}{W_f} + \arctan\frac{h_{RL} - h_{RR}}{W_r}\right),$$
$$\theta_k \approx \frac{1}{2}\left(\arctan\frac{(h_{FL} + h_{FR}) - (h_{RL} + h_{RR})}{L}\right).$$

These attitude estimates feed into the local terrain cost. For pitch, we use an asymmetric penalty that reflects the fact that steep uphill motion is more dangerous than moderate downhill motion:

$$J_\theta(x_k) = w_\theta^{up}\ \max(0, \theta_k)^2 + w_\theta^{down}\ \max(0, -\theta_k)^2, \quad w_\theta^{up} \gg w_\theta^{down}.$$

This makes negative pitch (downhill) less penalized than positive pitch (uphill), while still discouraging extreme descents.

We also include a soft belly-clearance term to discourage configurations where the chassis approaches the terrain:

$$J_{\text{clear}}(x_k) = w_c \,\text{softplus}(d_{\min} - c_{\text{clear}}(x_k))^2 ,$$

where $c_{\text{clear}}(x_k)$ is an estimate of clearance derived from the wheel heights and chassis geometry, and $d_{\min}$ is a desired safety margin.

Finally, these attitude and clearance costs are combined with the slope/roughness-based traversability score $T(x_k)$ from (1) to define the local terrain cost used in the MPC objective:

$$c_{\text{terrain}}(x_k) = \alpha\, T(x_k) + \beta\, J_\theta(x_k) + \gamma\, J_{\text{clear}}(x_k),$$

with nonnegative weights α, β, γ tuning the relative importance of global traversability, attitude safety, and clearance. For each candidate motion primitive along the MPC horizon, these quantities are evaluated from the elevation map, so the optimizer effectively accumulates the terrain cost of "how the ground will look" along the predicted base trajectory and can, for example, favor paths with gentler uphill slopes over more aggressive ascents, even if both satisfy the kinematic constraints.

This structure naturally integrates ARA* outputs, traversability fields, and hardware constraints. While the MPC is still in final code integration and tuning, the theoretical and implementation framework is firmly established and represents a meaningful upgrade over the original rubric expectation, since it moves beyond "stabilization control" into constrained optimal decision-making.

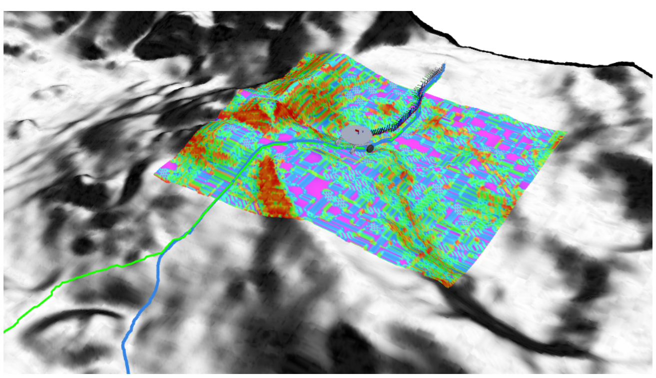

Fig. 12: Local traversability map feeding MPC controller.

VIII. Hardware Progress and Rubric Alignment

One of the clearest grading rubric expectations was the development of a robust, functioning physical platform capable of supporting meaningful autonomy experiments. In that regard, this semester represented both a significant expansion of scope and a partial deferral of final integration. Instead of making incremental fixes to the existing platform, we made the strategic decision to effectively rebuild HAMR 3.0 by leveraging lessons learned from GMU's COMPA and from our Vicon testing. We redesigned the entire base architecture, re-CADed all structural components, and produced a significantly more rigid chassis with improved mechanical integrity, better load distribution, and increased modularity. We also redesigned the turret–gimbal stack with a cleaner mechanical interface and stronger rails to support future sensors and payloads. In parallel, the team undertook development of spherical casters to replace the swivel casters identified experimentally as a major source of instability. As of the end of the semester, the robot is physically assembled with a completed mechanical structure, pending only the resolution of a faulty gimbal motor and final electronics installation before full control integration and outdoor testing.

It is also important to reflect on my role in this part of the project. While I was not the primary CAD designer for the hardware itself, I was deeply involved in leading the conceptual and architectural direction of the redesign. I coordinated priorities, helped define requirements, ensured that mechanical decisions remained consistent with our control and autonomy objectives, and guided the hardware team so that the final platform would genuinely support the traversability, MPC, and holonomic capabilities developed in the rest of this work. This was therefore a major learning experience in technical leadership, design coordination, and project management rather than purely mechanical execution, and it was essential to ensuring that the physical robot matches the system-level ambitions of HAMR 3.0.

Although the failure of one of the gimbal motors prevented final stabilization validation this semester, this limitation was not due to lack of technical progress or planning, but rather a late-stage component issue. All other subsystems—including mechanical readiness, structural design, and integration pathway—are in place. Therefore, even if the hardware is not yet fully "mission complete," I believe this semester strongly satisfies the rubric intent: we now have a significantly more capable, better engineered, and nearly deployment-ready HAMR 3.0 platform, with the remaining work consisting primarily of final integration rather than conceptual redesign.

IX. Software Architecture and Documentation

Beyond theoretical, simulation, and hardware contributions, a large part of making a robotics system credible lies in software structure, maintainability, and knowledge transfer. This semester included major effort in documenting the ROS 2 autonomy stack, including node architecture, launch organization, data flows, controller APIs, and simulation reproduction steps. This not only ensures that work is reusable by future students, but it also strengthens reliability since well-structured systems fail less and evolve more naturally. On the embedded side, the documentation effort led primarily by Kartik and Kai but also supported by me ensured that motor control tuning data, sensor calibration pipelines, and firmware processes are clear, repeatable, and transferable. This documentation fulfills the rubric expectation related to "system readiness and research maturity," extending the value of this independent study beyond a single semester.

X. Learning Outcomes

From an academic and engineering development standpoint, this semester provided extensive experience in perception-driven autonomy, off-road robotic reasoning, planning and controls formulation, structured simulation development, and research collaboration. The most meaningful personal outcome was learning how to intelligently pivot a research agenda: instead of rigidly following the initial rubric plan, I shifted priorities when better research opportunities emerged as we collaborated with GMU, while still maintaining alignment with the evaluation framework. The work completed this semester demonstrates not only technical competency, but also research judgment and system-level thinking.

XI. Status, Forward Work, and Conclusion

At the end of the semester, HAMR 3.0 stands as a significantly more mature platform than at its inception. The simulation environment is scientifically grounded and deeply useful. The traversability pipeline is functional and thoughtfully designed. The ARA* planner is integrated, the MPC controller framework is formed, and the physical robot is nearly deployment-ready pending final adjustments. Forward work will focus on final integration, outdoor testing, real-time traversability mapping, and full autonomy deployment.

Overall, while the project trajectory deviated deliberately from the exact original grading rubric wording, every adjustment strengthened the academic quality and practical relevance of the work. In many places, the outcomes significantly exceed the depth originally planned. HAMR 3.0 is on its way to become a credible off-road holonomic robotics research platform with a rich foundation to build upon next semester and beyond.

XII. Conclusion

This semester successfully advanced HAMR 3.0 into a mature research platform with credible simulation, validated experimental grounding, and a strong forward path toward off-road holonomic autonomy.

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